TOPIC 3: BONDING, PART 11

GRADE 11

SEMESTER 1/2020

Checklist

✓

3B Covalent bonding

3.10 Understand that covalent bonding is the strong electrostatic attraction between two nuclei and the shared a pair of electrons between them, based on the evidence:

I. The physical properties of giant atomic structure.

II. Electron density molecule of simple molecules

3.11 Be able to draw dot and cross diagrams to show electrons in covalent substances including:

I. Molecules with single, double and triple bonds

II. Species with dative covalent (co-ordinate) bonds, including Al_2Cl_6 and the ammonium ion

3.12 Be able to describe the different structures formed by giant lattices of carbon atoms, including graphite, diamond and graphene, and discuss the applications of each.

3.13 Understand the meaning of the term "electronegativity" as applied to atoms in a covalent bond.

3.14 know that ionic and covalent bonding are the extremes of continuum of bonding type and be able to explain this in terms of electronegativity differences, leading to bond polarity in bonds and molecule, and to ionic bonding if the electronegativity is large enough.

3.15 Be able to distinguish between polar bonds and polar molecules and predict whether or not a given molecule Is likely to be polar.

Further suggested practical

Determine the effect of electrostatic force on jets of liquids (water, ethanol and cyclohexane) and use the results to determine whether the molecules are polar or non-polar.

3C Shapes of molecules

3.16 Understand the principles of the electron pair-repulsion theory, used to interpret and predict the shapes of simple molecules and ions.

3.17 Understand the terms "bond length" and "bond angle"

3.18 Know and be able to explain the shapes of, and bond angles in $BeCl_2$, BCl_3, CH_4, NH_3, NH_4+, H_2O, CO_2, gaseous PCl_5, SF_6 and C_2H_4

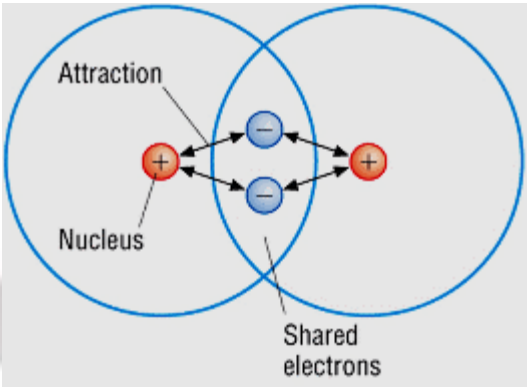
3.19 Be able to apply the electron-pair repulsion theory to predict the shapes of, and bond angles in, molecules and ions analogous to those in 3.18

3B Covalent Bonding

SPECIFICATION OBJECTIVE 3.10

The Covalent Bond

- The covalent bond is the ***strong electrostatic attraction between the two nuclei of the bonded atoms and the shared pair of electrons.***



- When a covalent bond is formed, two atomic orbitals overlap and a molecular orbital is formed.
- Like atomic orbitals, a molecular orbital can only contain two electrons.
- Overlap of atomic orbitals is thus only possible if both orbitals contain only one electron (normal covalent bond), or if one is full and the other is empty (dative covalent bond).
- Atoms can share more than one pair of electrons and form double or triple covalent bonds:
  - I. a **single bond** results from one shared electron pairs – e.g. in chlorine, Cl_2, Cl—Cl
  - II. a **double bond** results from two shared electron pairs – e.g. in oxygen, O_2, O=O
  - III. a **triple bond** results from three shared electron pairs – e.g. in nitrogen, N_2, N≡N.
- Depending on the pattern of overlapping of atomic orbitals, the covalent bond can be sigma (σ) or pi (π) bond.

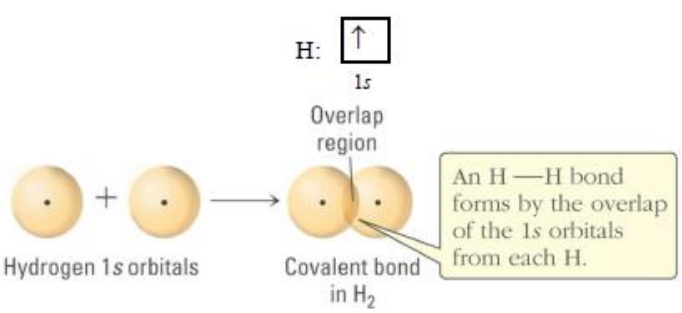
I. Single covalent bond or the sigma (σ) bond

- For a single covalent bond, an orbital with one electron should linearly overlap with an orbital with one electron of the other atom.
- There can only be one sigma bond between two atoms.

Example:

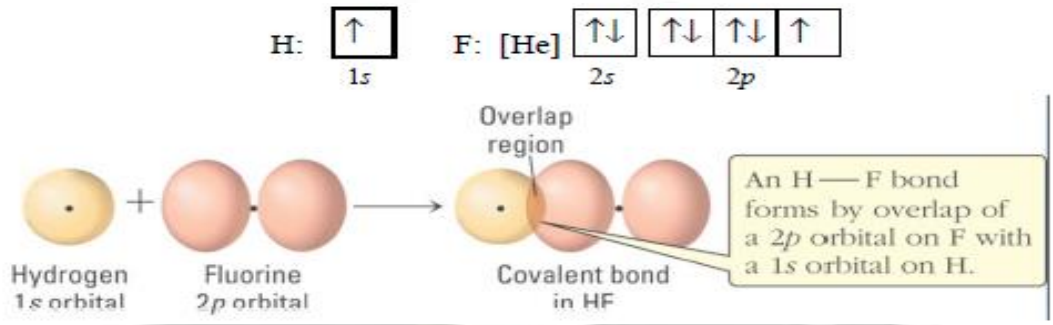
- **A. Linear overlap of two `s' orbitals (H_2 molecule)**

Example: formation of hydrogen, H_2



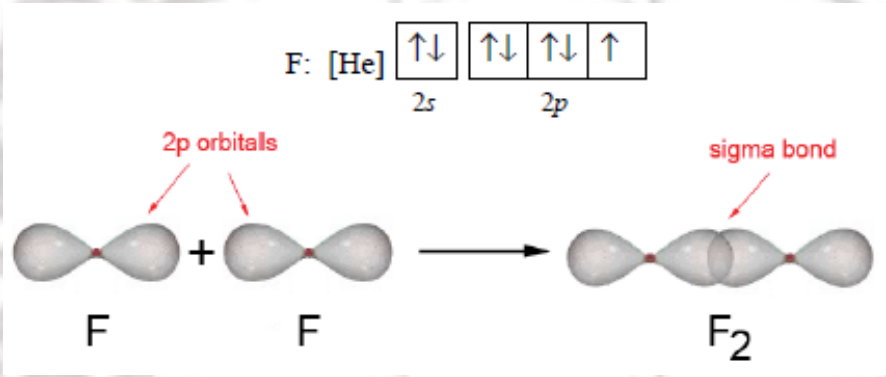
- B. Linear overlap of an `s' orbital and a `p' orbital (HBr molecule)

Example: *Formation of HF*

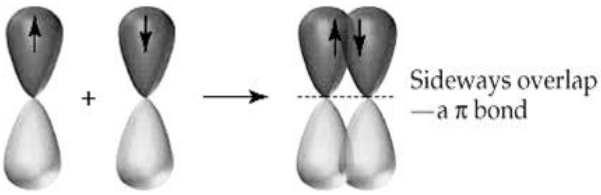


- C. Linear overlap of two `p' orbitals (F_2 molecule)

Example: *formation of fluorine, F_2*

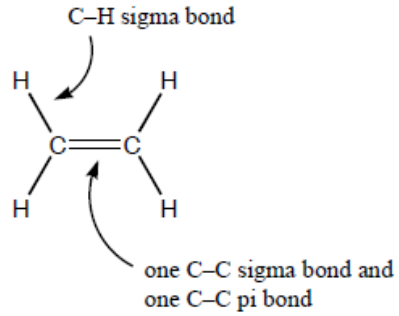


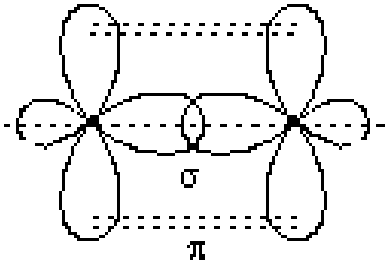
II. Double covalent bond or one sigma (σ) bond together with a pi (π) bond



- When a double bond is formed between two atoms, one bond is a sigma bond and the other is a pi bond.
- Pi bond is formed when two p orbitals overlap laterally.
- Each p orbital has two lobes one above and one below the plane.
- when the two orbital overlap to form a bond where the bonding region is above and below the two nuclei and not directly between them.
- The sideways overlap of two p orbitals to form a pi bond is less effective and results in a weaker bond than the sigma bond formed by direct, head-on overlap of atomic orbitals.

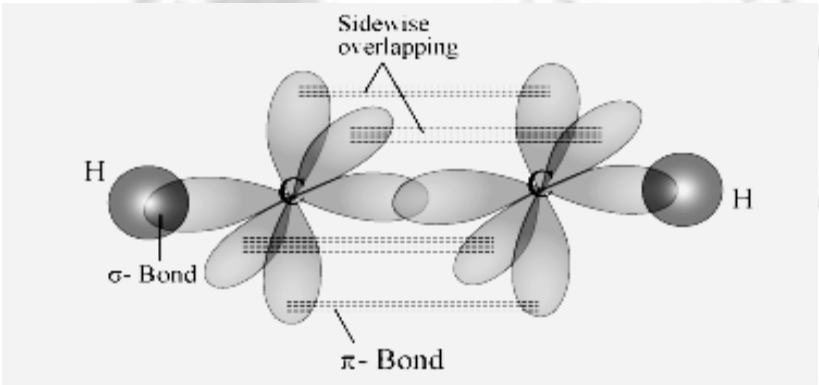
Example: ethane, C_2H_4

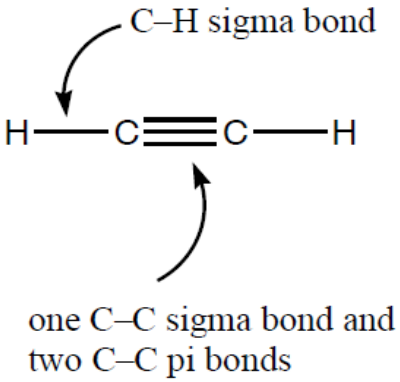




III. Triple covalent bond or one sigma (σ) bond together with two pi (π) bonds

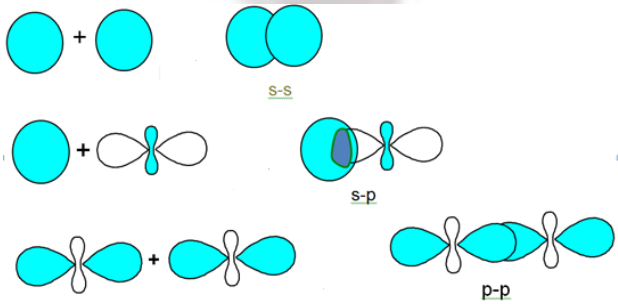
- When a triple bond is formed between two atoms, one bond is a sigma bond and the other two are pi bonds.
- Example: *ethyne, H-C≡C-H*



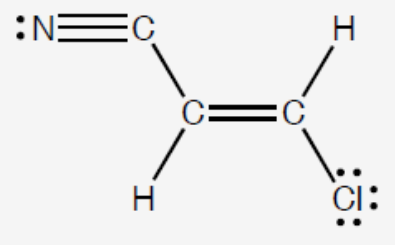


EXERCISE 1

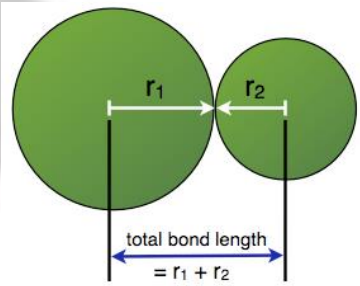
Question 1: State the type of bonding shown in each.



Question 2: How many sigma and pi bonds are in the following molecule?



Bond Length and bond strength



- Bond length is the distance between the nuclei of two atoms that are covalently bonded together.
- Strength of covalent bond is measured by the amount of energy required to break one mole of the bond in gaseous state.

BOND	BOND LENGTH / nm	BOND STRENGTH / kJ mol^{-1}
Cl–Cl	0.199	242
Br–Br	0.228	193
I–I	0.267	151
C–C	0.154	347
C=C	0.134	612
C≡C	0.120	838

- Shorter bonds have greater bond strength due to increased electrostatic attraction between the two nuclei and the electrons in the overlapping atomic orbitals.

CHECKPOINT 1

Do the check point questions on page 74 from text book.

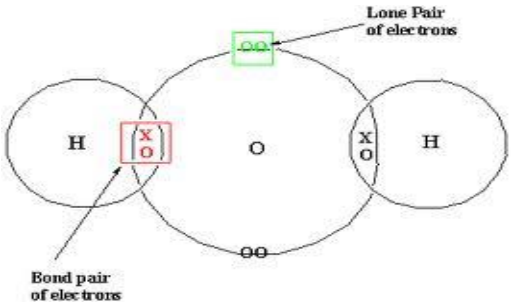
3B Dot and Cross diagram for covalent compounds in discrete molecules

SPECIFICATION OBJECTIVE 3.11,i

A discrete molecule is an electrically neutral group of two or more atoms held together by covalent bonds.

When drawing the arrangement of electrons in a molecule using dot and cross diagrams

- Use dots for electrons from one of the atoms and a 'cross' for the electrons from the other atom.
- If there are more than two types of atom use additional symbols such as a small circle or a small triangle.
- Draw the outer electrons in pairs to emphasize the number of bond pairs and the number of lone pairs (non- bonded).



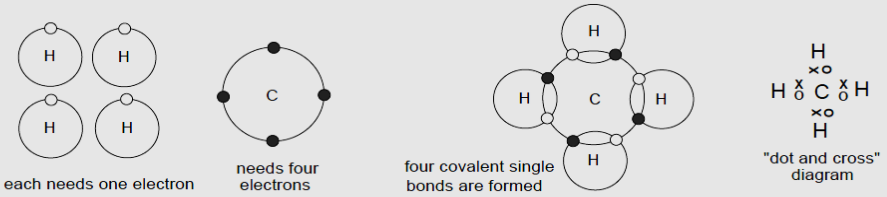
Lone pair of electrons: An electron pair that is part of an atom's valence electron shell but not involved in bonding

Worked Example

WATER



METHANE



EXERCISE 2

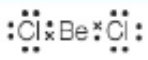
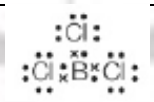
Draw the dot/cross diagram for the following molecules and displayed formula.

a) HCl	b) CO_2
b) O_2	d) PCl_5

Displayed formula (full structural formula) : shows each bonding pair with a line between two atoms.

The octet rule

- In most cases, covalent bonded molecules have the same number of electrons as the outer shell of noble gas. Mostly that is 8 electrons referred as "octet rule".
- But this is not always true in some cases like $BeCl_2$, BCl_3, PCl_5, SF_6.

Compound	Dot and cross diagram	Number of electrons around central atom
Beryllium chloride, $BeCl_2$		4
Boron trichloride, BCl_3		6
Phosphorous(V) Chloride (PCl_5)		10

Sulfur hexafluoride (SF_6)		12

CHECKPOINT 2

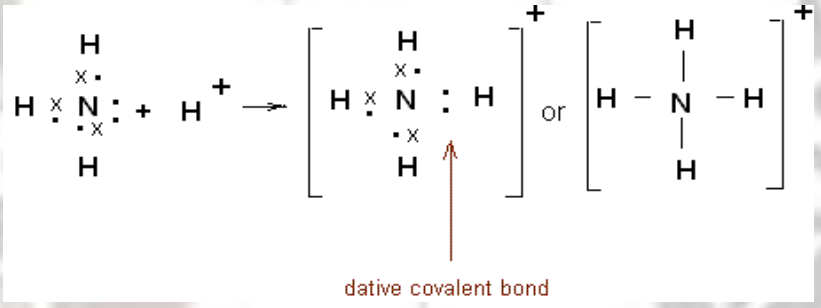
- Read page 79 of text book.
- Do the check point questions on page 79 in your text book.

3B Dative covalent bonds

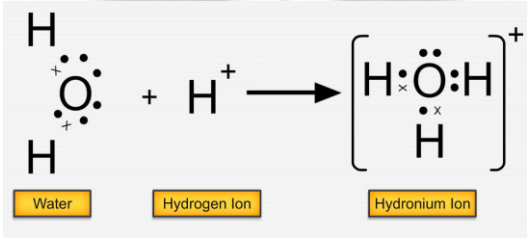
SPECIFICATION OBJECTIVE 3.11,ii

- Dative covalent bonds (coordinate bond) are formed when both the shared electrons come from just one of the atoms.
- It is formed when an orbital of one atom containing two electrons overlap with an empty orbital of the other atom. The basic requirement will be an atom with a lone pair of electrons (non-bonded) and the other atom having an orbital with no electrons.

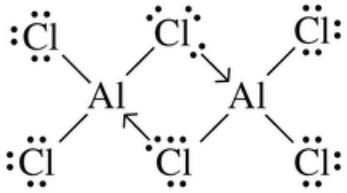
e.g: Formation of the dative covalent bond in the ammonium ion.

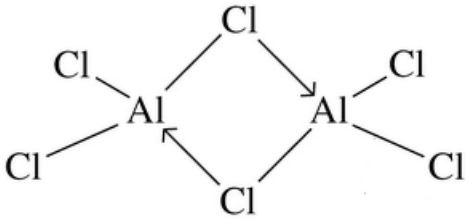


eg: Formation of the dative covalent bond in the hydrozonium ion.



eg: Formation of the dative covalent bond in aluminium chloride dimer





EXERCISE 3

Question 1: One way of describing the bonding in a molecule of carbon monoxide is to state that it contains two covalent bonds and one dative covalent bond. Using this description, draw a dot and cross diagram for a molecule of carbon monoxide?

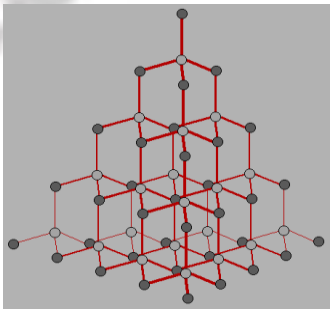
Question 2. Draw a dot and cross diagram for the $AlCl_4^-$ and $NH_3.BF_3$ and identify the dative covalent bond.

3B The evidence for covalent bonds

SPECIFICATION OBJECTIVE 3.10

Physical properties

The physical properties of **giant atomic structures** such as diamond and graphite provide evidence for the strong electrostatic attraction in covalent bonding. Giant atomic structures are also known as **giant molecular structures**.

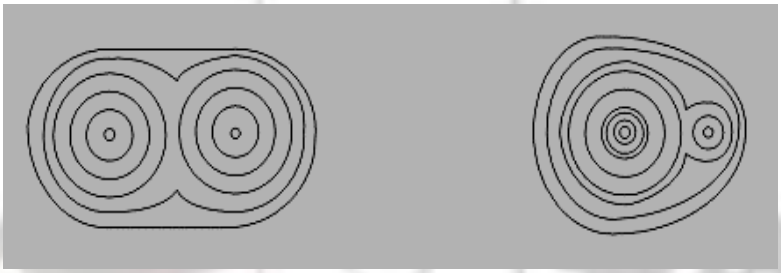


- They are very hard and have high melting temperatures. This is because the structures are made up of a large number of strong covalent bonds, all of which need to be broken if the atoms are to be separated.

- Do not conduct electricity as they have no mobile ions or electrons BUT **Graphite conducts electricity.** Each atom only uses three of its outer shell electrons for bonding to other carbon atoms and the remaining electron can move through layers allowing the conduction of electricity.
- Insoluble in water and other solvents.

Electron density maps

Electron density maps show high electron density between atoms that are covalently bonded.



Cl_2 HCl

SPECIFICATION OBJECTIVE 3.12

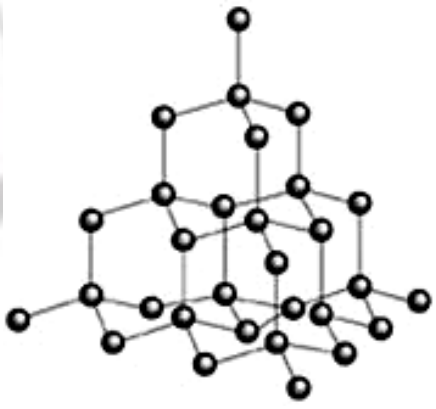
3B Giant covalent lattices

Giant covalent lattices are sometimes called network of covalent lattices. They consist of a giant network of atoms linked to each other by covalent bonds

The most common giant covalent substances are:

- Diamond
- Graphite
- Graphene

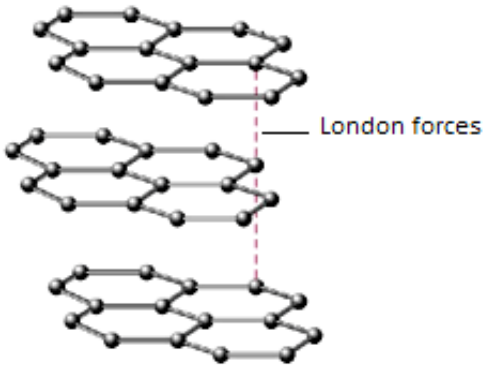
Diamond:



- Composed of interlocking tetrahedral carbon atoms.
- Each carbon atom forms four sigma bonds to four other carbon atoms in a giant three dimensional tetrahedral arrangements.
- Diamond is extremely hard because of the very strong C-C bonding throughout the structure.
- It also has very high melting temperature because a great number of strong C-C bonds have to be broken in order to melt it. This requires large amount of energy.
- Uses are based on its exceptional hardness. Used in drilling, cutting and grinding and as bearings.
- Can be made artificially by subjecting graphite to high temperatures and pressures.

Graphite

- Composed of planes of trigonal carbon atoms joined in a honeycomb pattern.
- Each carbon atom is bonded to three other carbon atoms at 120°.
- Weak London forces (van der Waals forces) exist between layers.
- Used as a lubricant - it is slippery as the layers' slide over each other.
- Used as 'lead' in pencils, as an electrical conductor and electrode material in dry cells and in nuclear reactors.

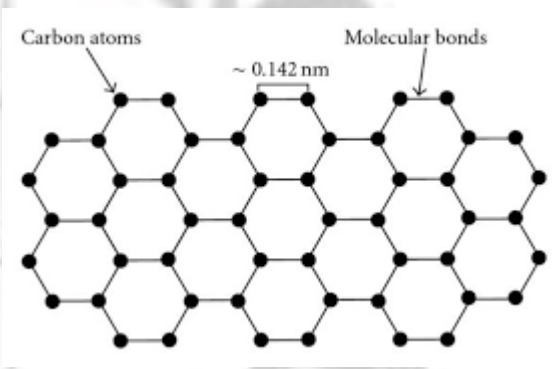


Graphite

- Can conduct electricity as there are free electrons (one in each atom).

Graphene

- Graphene is pure carbon in the form of very thin sheet, one atom thick.
- The carbon atoms are arranged in a honeycomb-like pattern.
- The carbon atoms are bonded in exactly the same way as in graphite and it can, therefore be described as one atom thick layer of graphite.



- It is transparent, flexible, highly conductive.
- A sheet of graphene can be rolled into a ball to produce fullerene molecules.
- A sheet of graphene can be rolled into a cylinder to produce carbon nanotubes.
- Graphene is very strong, and so can be used to enhance the strength of other materials. Research have demonstrated that adding even a trade amount of graphene to plastics, metals or other materials can make these materials much stronger.
- Graphene is highly conductive, hence it could be useful in both microelectronics (for example to make LED lighting more efficient and longer lasting) and also in larger applications - for example thermal foils for mobile devices.

CHECKPOINT 3

- Read page 88, 98, 90 & 91 of text book.
- Do the check point questions on page 91 in your text book.

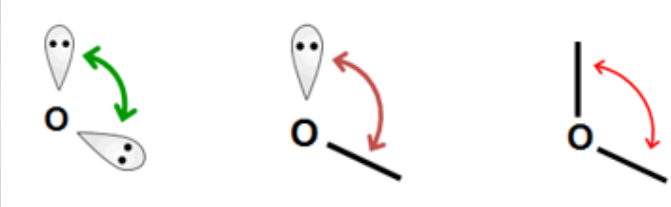
3C Shapes of simple molecules and ions

SPECIFICATION OBJECTIVE 3.16, 3.17, 3.18 and 3.19

Valence Shell Electron pair repulsion theory

This is used to predict the shapes of simple molecules and ions by considering the repulsions between pairs of electrons (lone pair and bond pair) within the molecule. It states that,

- *The shape of a molecule or ion is largely determined by the* ***total number of electron pairs*** *in valance shell of the central atom.*
- *The electron pairs arrange themselves as far apart as possible to* ***minimize the repulsion****(maximum separation).*
- ***Lone pair –lone pair repulsion > lone pair – bond pair repulsion> bond pair-bond pair repulsion***



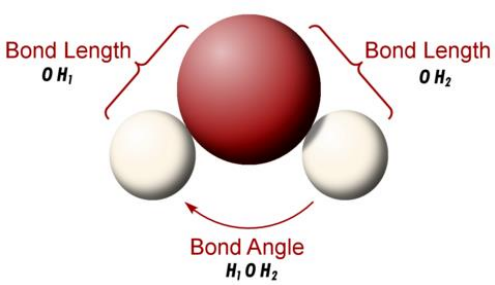
"The shape adopted is the one which keeps repulsive forces to a minimum"

To determine the shape, count up the number of covalent bond pairs and lone pairs around the central atom and work out the shape which keeps the bonds as far apart as possible.

Some useful terminologies;

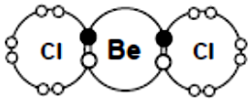
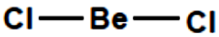
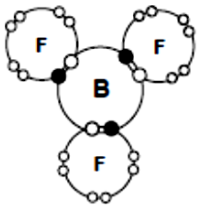
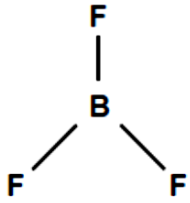
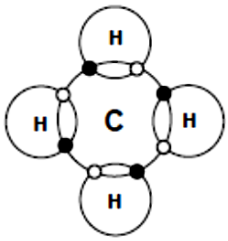
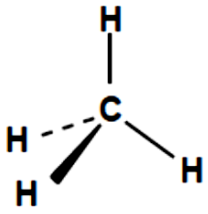
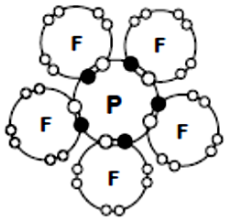
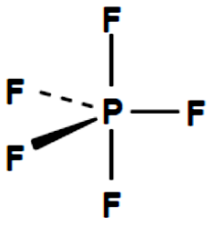
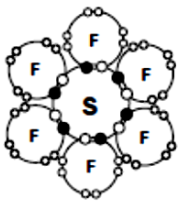
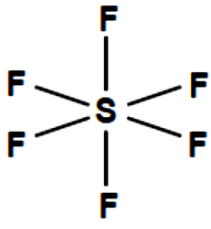
Bond length is the average distance between the centers of the nuclei of two bonded atoms in a molecule.

Bond angle: molecules made up of three or more atoms, the average angle between the bonded orbitals (i.e., between the two covalent bonds) is known as bond angle θ.



I. Species without lone pairs

Only bond pair repulsions occur and the basic shapes are regular. Therefore, the shape of the molecules depends on the number of bond pairs around the central atom.

MOLECULE	STRUCTURE		BOND PAIRS	BOND ANGLE(S)	SHAPE
$BeCl_2$			2	180°	Linear
BF_3			3	120°	Trigonal planar
CH_4			4	109.5°	Tetrahedral
PF_5			5	120° 90°	Trigonal bipyramidal
SF_6			6	90°	Octahedral

CHECKPOINT 4

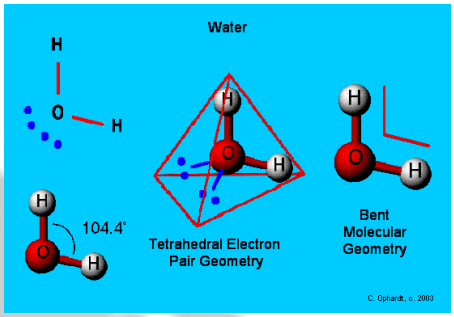
- Read page 83 of text book.
- Do the check point questions on page 83 in your text book.

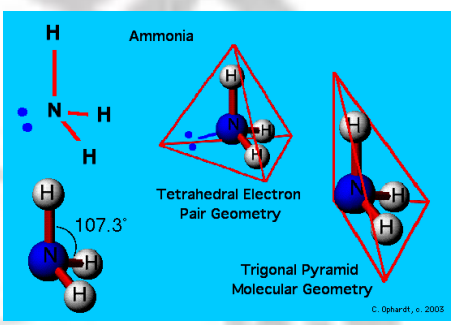
II. Species with lone pairs

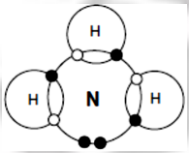
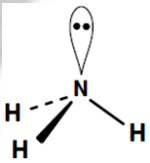
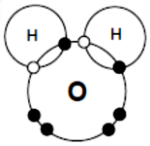
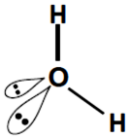
Lone pairs of electrons have a greater repulsive power than bond pairs so their presence will affect the angles of bonds as they push the bond pairs away. The order of repulsive power is…

lone pair – lone pair >lone pair – bond pair>bond pair – bond pair

- The resulting configuration is based on the number of electron pairs but the actual shape **does not include the lone pairs.**
- A water molecule is angular despite the fact that it has 4 electron pairs around oxygen. Two of the pairs are lone pairs and are "invisible".
- Lone pairs distort the shape of molecule and reduce the bond angle. The following examples show the effect due to the presence of lone pair of electrons. Each lone pair decreases the bond angle by**2.5^0**.

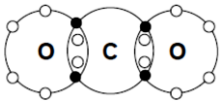
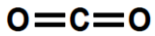
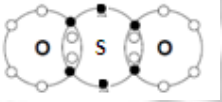
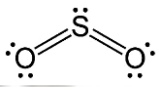




Worked Examples			
Ammonia	**3 bond pairs** and **1 lone pair** (total = 4 pairs) The shape is based on a tetrahedron. As the lone pair-bond pair repulsions are greater than bond pair-bond pair repulsions the H-N-H bond angle is reduced from 109.5° to **107°**. :- shape is **PYRAMIDAL**		
Water	**2 bond pairs** and **2 lone pairs** (total = 4 pairs) The shape is based on a tetrahedron. The extra lone pair-lone pair repulsion pushes the H-O-H bond angle down further to **104.5°**. :- shape is **ANGULAR/ BENT**		

III. Species with double bonds

For repulsive purposes, **double bonds act like single bonds.**

Worked Examples			
Carbon dioxide	**2 double bond pairs** and **no lone pairs** Since the double bonds act like single bonds, the shape will be based on two bond pairs repelling each other. The bond angle is **180°**. :- shape is **LINEAR**		
Sulphur dioxide	**2 double bond pairs** and **one lone pair** Since the double bonds act like single bonds, the shape will be based on two bond pairs and one lone pair repelling each other. The bond angle is **117.5°**. :- shape is **ANGULAR/ BENT**		

EXERCISE 4

Draw the structures of the following molecules indicating the bond angles and the shape of molecules.

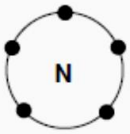
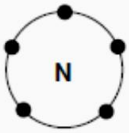
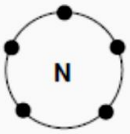
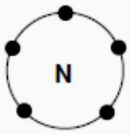
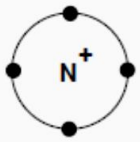
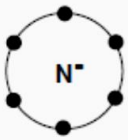
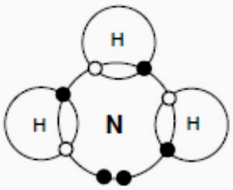
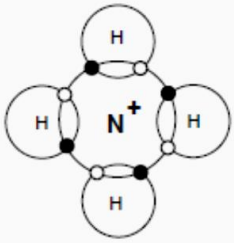
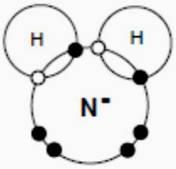
MOLECULE	ELECTRON PAIRS (Bond pairs and lone pairs)	SHAPE (including bond angle)	NAME OF SHAPE
BF_3			

MOLECULE	ELECTRON PAIRS (Bond pairs and lone pairs)	SHAPE (including bond angle)	NAME OF SHAPE
BeI_2			
CH_2Cl_2			
PH_3			
H_2S			

MOLECULE	ELECTRON PAIRS (Bond pairs and lone pairs)	SHAPE (including bond angle)	NAME OF SHAPE
PCl_3			
OF_2			
PBr_5			
SCl_6			

IV. Shape of simple ions

Shapes can be worked out according to the method shown. It allows you to predict the shape but, in some cases, not the true nature of the bonding.

	NH_3	NH_4^+	NH_2^-
Draw out the OUTER electronic configuration of the central atom.			
If the species is an ion ... Add one electron for each negative charge or remove one electron for each positive charge			
Pair up the electrons of the central species with those of the atom(s) surrounding it. Count the electron pairs.			
ELECTRON PAIRS	**BOND PAIRS 3** **LONE PAIRS 1**	**BOND PAIRS 4** **LONE PAIRS 0**	**BOND PAIRS 2** **LONE PAIRS 2**
SHAPE	**PYRAMIDAL**	**TETRAHEDRAL**	**ANGULAR**

CHECKPOINT 5

- Read page 82 & 83 of text book.
- Do the check point questions on page 83 in your text book.

EXERCISE 5

Draw the structures of the following ions indicating the bond angles and the shape of molecules and ions.

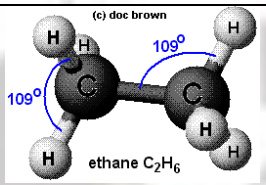
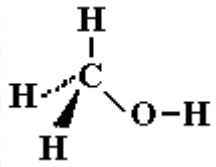
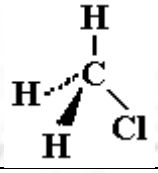
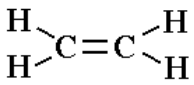
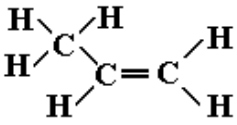
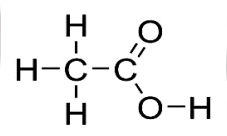
ION	Draw out the OUTER electronic configuration of the central atom.	Electron pairs (Bond pairs and lone pairs)	Structure (shape and bond angle)
AlH_4^-			
ICl_2^+			
PF_6^-			
H_3O^+			

V. Shapes of organic molecules

These molecules are not given a particular shape name; however, the 3D spatial arrangement is expected.

EXERCISE 6

Write the bond angles of the following organic molecules.

MOLECULE	STRUCTURE	BOND ANGLES
Ethane		H-C-H= C-C-H =
Methanol		H-C-H = H-C-O = C-O-H=
chloromethane		H-C-H = H-C-Cl =
Ethene		H-C-H = C =C-H =
propene		H-C-H (in methyl group)= C-C=C = H-C-H (in alkene) =
Ethanoic acid		H-C-H = H-C-C = C-C-O = C-O-H =

3B Electronegativity and Bond polarity

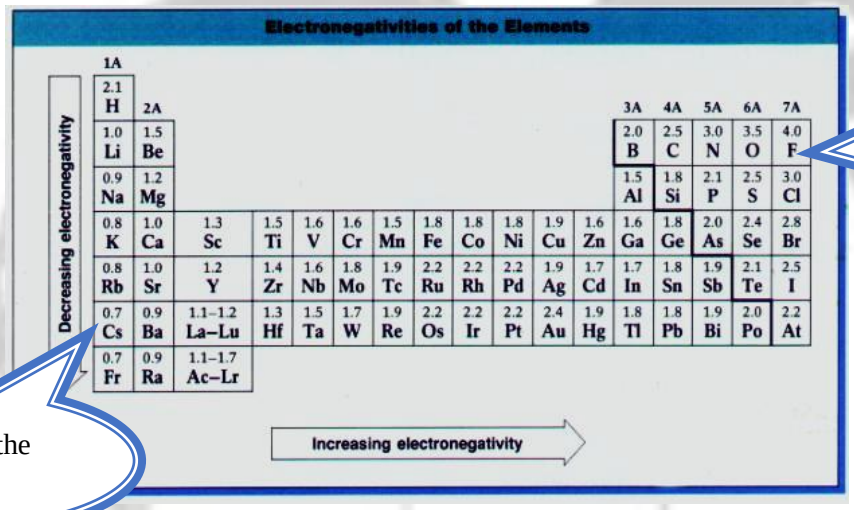
SPECIFICATION OBJECTIVE 3.13 and 3.14

Most bonds aren't purely ionic or purely covalent but somewhere in between (intermediate bonding). Electronegativity is the main factor which leads to these intermediate bonding.

Electronegativity

'The ability of an atom to attract the bonding pair of electrons in a covalent bond to itself.'

- Electronegativity is usually measured using the Pauling scale
- Fluorine is the most electronegative element according to this scale (it's given a value of 4.0)



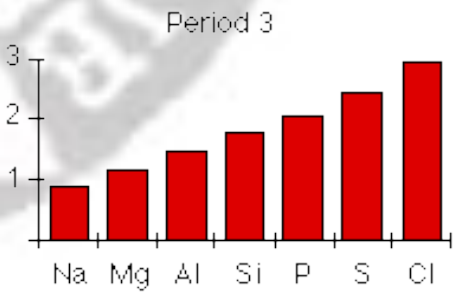
Fluorine has the highest value (4.0)

Cesium has the lowest value

Trends in electronegativity

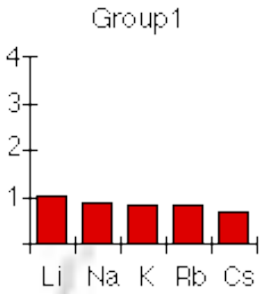
Electronegativity increases across the periods as

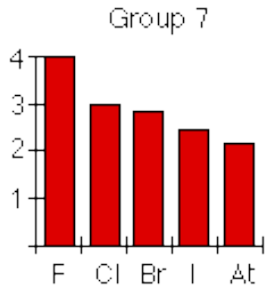
- nuclear charge increases across the period
- outer electrons are in the same shell
- there is no extra shielding
- therefore, there is more attraction between the nucleus and the shared pair of electrons



Electronegativity decreases down the groups as

- The outer electrons are further away from the nucleus
- There is increased shielding
- therefore, there is less attraction between the nucleus and the shared pair of electrons





Distribution of electron density in a covalent bond

Non-polar bond

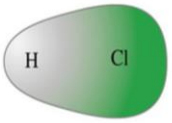
- Similar atoms have the same electronegativity
- They will both pull on the electrons to the same extent and hence the electrons will be equally shared
- The electron density will be symmetrical
- Therefore, it is a **non- polar** covalent bond or pure covalent bond.

$Cl-Cl$ $O=O$ $N\equiv N$

Polar covalent bond

The covalent bonds in which the bonded atoms differ in the electronegativities will share the electron pair **unequally**, giving rise to partial negative charge (δ^-) on the atom having more electronegativity and partial positive charge (δ^+) on the atom having less electronegativity. The resulting covalent bond is called a **polar covalent bond**.

- Different atoms have different electronegativities.
- Atom with the higher electronegativity will pull the electron pair closer to its end.
- The electron density will be unsymmetrical, one end will be slightly more negative, delta negative (δ^-) than average and the other end will be slightly less negative, or more positive, delta positive (δ^+).
- A dipole is induced and the bond is said to be **polar**
- The greater the electronegativity difference, the greater the bond polarity.

	**HCl polar bond showing the unequal sharing of a cloud of electron density.**
$\overset{\delta+}{H}-\overset{\delta-}{Cl}$	**HCl polar bond showing the partial (δ+) change on hydrogen and the partial (δ -) change on chlorine.**
$\overset{\longrightarrow}{H-Cl}$	**HCl polar bond showing the direction of the dipole with an arrow pointing toward the more negative atom. The + on the opposite end also reminds us which atom is more positive.**

***Note**: A polar covalent bond can be represented using an arrow to show the direction of electron drift.*

Using Electronegativity to Identify Ionic, Covalent, and Polar Covalent

Electronegativity values are useful in determining if a bond is to be classified as nonpolar covalent, polar covalent or ionic.

- The ionic and covalent bonding are two extremes of bonding and in practice all degrees of intermediate bonding occur which have both ionic and covalent characteristics.
- A covalent bond that has some degree of covalent character is a polar covalent bond.
- Therefore, the degree of ionic and covalent character depends on the electronegativity difference between the two atoms.
- It is somewhat artificial to classify bonds based on the differences in the electronegativities (Δ_X) of the two atoms. However, these ranges can be used to do so:

$\Delta_X \geq 1.7$	Ionic (predominantly ionic with some covalent character)	(symbolized as A^+ and Z^-)
$1.7 > \Delta_X > 0$	Polar Covalent (predominantly covalent with some ionic character)	(symbolized as $A^{\delta+}$ and $Z^{\delta-}$)
$\Delta_X = 0$	Pure Covalent	(no charges)

As a rule, when the difference between the electronegativities of two elements is greater than 1.7, we assume that the bond between atoms of these elements is **ionic.**

- Example

NaCl		*LiF*		$SrBr_2$	
Cl	*EN* = 3.0	F	*EN* = 4.0	Br	*EN* = 2.8
Na	*EN* = 0.9	Li	*EN* = 1.0	Sr	*EN* = 1.0
	---		---		---
	ΔEN = 2.1		ΔEN = 3.00		ΔEN = 1.8

when the difference between the electronegativities of two elements is same, the atoms share electrons, and the substance is pure *covalent*.

- Example: H_2 or Cl_2 molecules

As a rule, when the difference between the electronegativities of two elements is less than 1.7 and greater than zero, we assume that the bond between atoms of these elements as ***polar*****,** or ***polar covalent*****.**

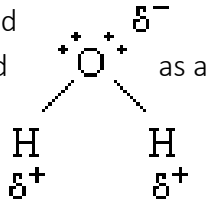
Example: water H_2O

Electronegativity of oxygen = 3.0

Electronegativity of hydrogen EN = 2.1

Difference in electronegativity = 0.9

Water is neither purely ionic nor purely covalent. It doesn't contain positive and negative ions and the electrons are not shared equally. Water is best described as a polar compound. One end, or pole, of the molecule has a partial positive charge (δ+), and the other end has a partial negative charge (δ-).



EXERCISE 7

Question 1: In each pair of bonds, indicate the more polar bond and use an arrow to show the direction of polarity in each bond.

a) C—O and C—N	b) B—O and B—S
c) P—Br and P—Cl	d) B—F and B—I

Question 2: Classify each bond (*ionic, polar covalent or pure covalent*) and indicate full or partial charges, if any.

a) Na—Cl	b) Cs—F
c) C—H	d) Mg—H
e) Cl—Cl	f) C—O

CHECKPOINT 6

- Read page 77 of text book.
- Do the check point questions on page 77 in your text book.

3B Polarity of molecules

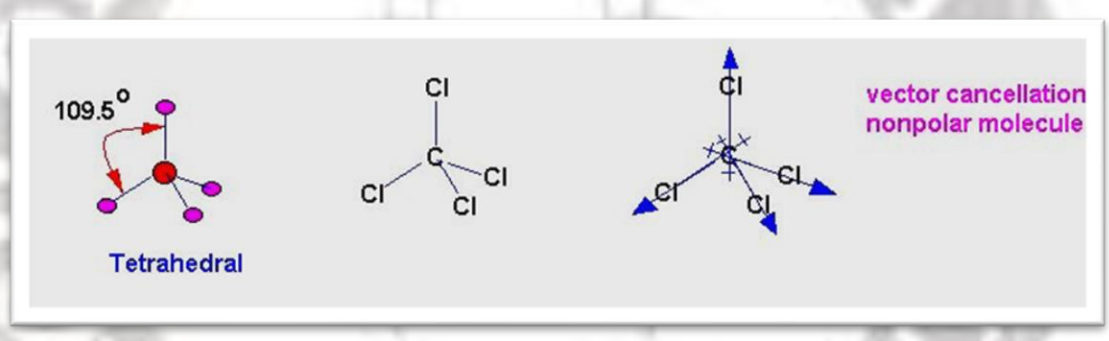
SPECIFICATION OBJECTIVE 3.15

- Polarity is a vector quantity that has both magnitude and direction.
- Some molecules that contain polar bonds are polar (the molecule has a permanent dipole).
- Some molecules may not be polar even if they contain polar bonds.
- the molecules will be polar if they have a NET DIPOLE MOMENT
- it is a bit like balanced forces

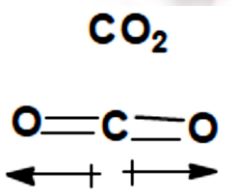
NON POLAR MOLECULE → • dipoles in bonds within the molecule 'cancel each other'

POLAR MOLECULE → • dipoles in bonds within the molecule do not 'cancel each other'

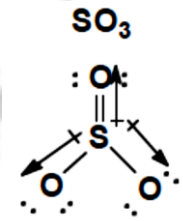
➢ In symmetrical molecules though the bonds are polar, the dipoles from each of the bonds cancel out resulting in no overall polarity. These molecules are non-polar.



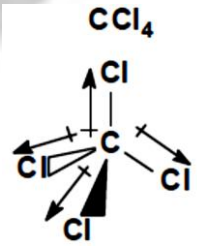
- Therefore, linear, trigonal planar, tetrahedral, trigonal bipyramidal and octahedral molecules are non- polar molecules (provided that similar atoms are bonded to the central atom).



Linear

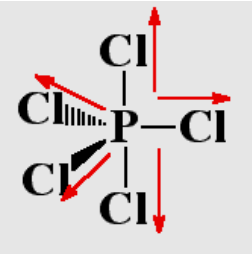


Trigonal planar



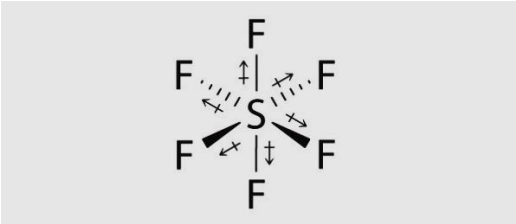
Tetrahedral

PCl_5



Trigonal bipyramidal

SF_6



Octahedral

Difference between polar bond and polar molecules

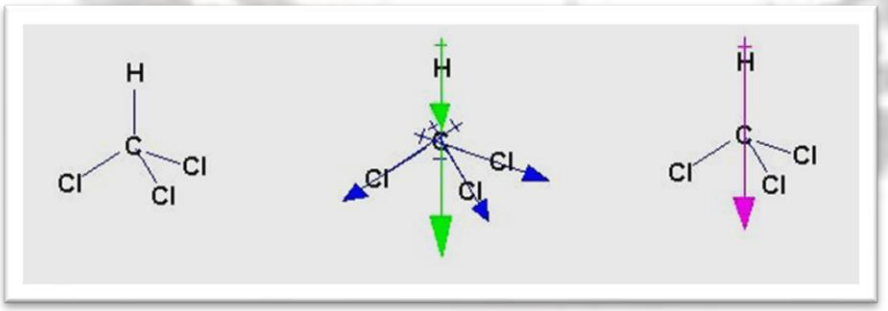
In a simple molecule like HCl, the bond is polar because the electronegativities of chlorine and hydrogen is large enough to assume that the bond is assumed to be polar. HCl contains only H-Cl bond, so the HCl molecule is also described as **polar**.

In symmetrical molecules though the bonds are polar, the dipoles from each of the bonds cancel out resulting in **no overall polarity**. These molecules are **non-polar**.

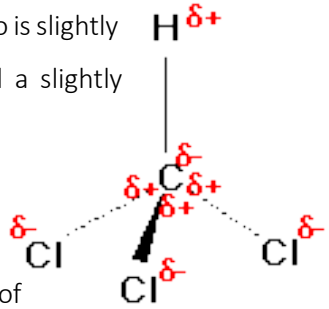
In tetrachloromethane, CCl_4, each bond is **polar**, however, the molecule as a whole, is **not polar** – In other words, it does not have an end (or a side) which is slightly negative and one which is slightly positive. There is no overall separation of charge from top to bottom, or from left to right.

- Molecules will **be polar** if bond polarities from each of the **bonds do not cancel out** in the molecule.

Example: Trichloromethane, $CHCl_3$

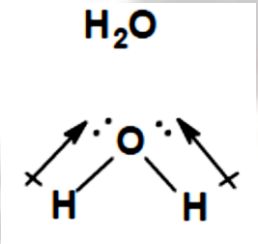


The hydrogen at the top of the molecule is less electronegative than carbon and so is slightly positive. This means that the molecule now has a slightly positive "top" and a slightly negative "bottom", and so is overall a **polar molecule**.

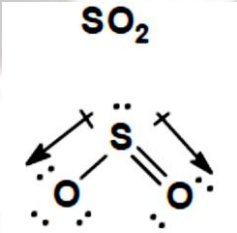


The electrons in the trichloromethane molecule are attracted towards one end of the molecule and this distortion of the electron cloud means that the centre of positive charge and the centre of the negative charge do not occur in the same place in the molecule.

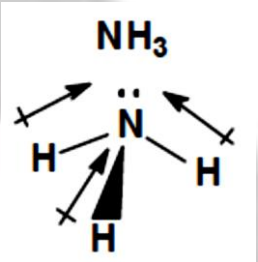
- Angular and pyramidal molecules are polar.



Angular



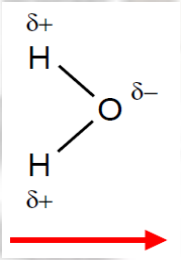
Angular



Pyramidal

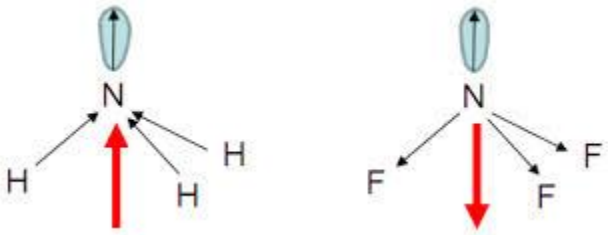
Example: water, H_2O

A Water molecule consists of **two Hydrogen atoms** covalently bonded to an **Oxygen atom**. Because oxygen is more electronegative than hydrogen, it has a **greater pull** on the **shared electrons**. This that the oxygen atom is **slightly negative (δ-)** (because of the closer electrons), and hydrogen is **slightly positive (δ+)**. Because the molecule is not symmetrical, it has an overall moment and therefore it is a **Polar Molecule**.



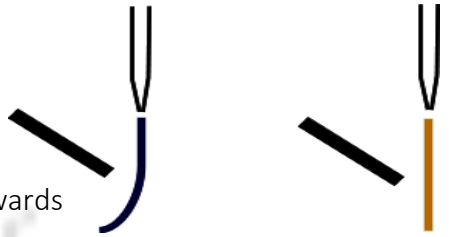
Exceptions

NF_3 has the same geometry as NH_3, but the molecule has very little polarity or considered as non-polar because the lone pair cancels out the polarity of N-F bonds.



Experiment to determine the polarity of molecules

- Place a liquid in a burette.
- Allow a narrow stream to run out.
- Place a charged rod next to the flow of the liquid.
- Polar molecules will be attracted and deflected towards the charged rod.



Polar Non-polar

EXERCISE 8

Complete the Table.

molecule	Shape of the molecule	Polarity of bond	Polarity of molecule
N_2			
NH_3			
OCl_2			
$SiCl_4$			

CHECKPOINT 7

- Read page 84 & 85 of text book.
- Do the check point questions on page 85 in your text book.

3E Structure and properties

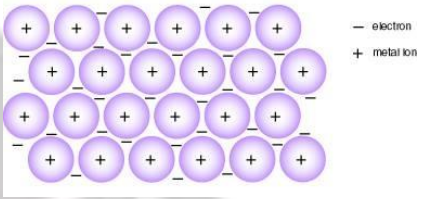
SPECIFICATION OBJECTIVE 3.12

The four types of solid structures covered in this topic are

1. Giant metallic lattices
2. Giant ionic lattices
3. Giant covalent lattices
4. Discrete (simple) molecular lattices

Metallic lattices

- Composed of a regular arrangement of positive metal (cations) ions surrounded by sea of mobile electrons



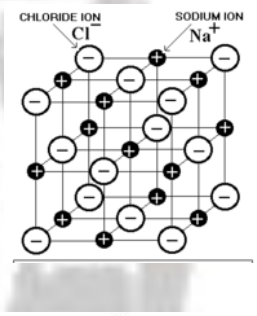
Properties of giant metallic lattices

- High melting and boiling temperatures
- Good thermal and electrical conductivity
- Malleability and ductility

Giant ionic lattices

- Composed of a regular arrangement of positive ions (cations) and negative ions (anions)

Properties of giant ionic lattices



- Fairly High melting and boiling temperatures
- Brittle
- Poor electrical conductivity when solid but conducts when molten or aqueous.
- Often soluble in water

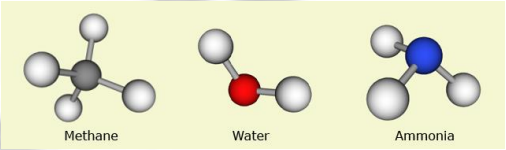
Giant covalent lattices



- Giant covalent lattices are also called **network covalent lattices**.
- Composed of a giant network of atoms linked to each other by strong covalent bonds

Molecular solids

- Are collections of molecules held together by intermolecular forces (IMFs). The solid structure is maintained by intermolecular forces rather than bonds (metallic, ionic or covalent).



Physical properties of molecular solids

In general, they have;

- Low melting and boiling temperatures
  In order to melt molecular solids energy is required to break the weak intermolecular forces, which lower than the energy required to break covalent bonds.

Types of bonding and structure

Bonding	Structure	Examples
Metallic	Giant lattice	Mg, Al, Cu, Zn, Na
Ionic	Giant lattice	NaCl, MgO, CsF
Covalent (including polar covalent)	Giant lattice Molecular Macromolecular	C(diamond), C(graphite), Si, SiO_2 (sand), BN H_2O, I_2, P_4, S_8, C_{60}, C_{12},$H_{22}O_{11}$(sucrose) Polymers eg; polythene, proteins, DNA

Summary of major properties of each type of structure

	Giant metallic	Giant ionic	Giant covalent	Simple molecular
Particles present	Positive ions and delocalised electrons	Positive and negative ions	atoms	Molecules
Type of bonding	Metallic	Ionic	Covalent	Covalent
Presence of intermolecular forces.	No	No	No	Yes
Melting and boiling temperatures	Fairly high to high	Fairly high to high	High to very high	Generally low
Electrical conductivity	Good when solid and molten	Non conductor when solid, conducts when molten and aqueous	Non conductor	Non conductor
Solubility in water	Insoluble unless metal reacts with water. Eg; Na, Li.	Generally soluble, exceptions are AgCl, AgBr, AgI and $BaSO_4$	Insoluble	Generally insoluble but may dissolve if hydrogen bonding is possible (eg: sucrose) or if substance reacts with water (eg Cl_2).

CHECKPOINT 8

- Read page 92 & 93 of text book.
- Do the check point questions on page 93 in your text book.